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Coupling assembly for coupling a rod to a bone anchoring element, polyaxial bone anchoring device and modular stabilization device

Abstract

A coupling assembly for coupling a rod to a bone anchoring element, and including a receiving part having a first end, a second end, and an accommodation space having an opening at the second end for inserting a head of the bone anchoring element, a pressure element insertable from the first end, positionable in the accommodation space, and configured to hold the head, the pressure element having at least one spring portion extending circumferentially around the pressure element and being radially compressible, and a rod receiving element separable from the pressure element, wherein in a first position, the spring portion is configured to engage an engagement structure at an inner wall of the receiving part to prevent movement of the pressure element towards the first end, and wherein the rod receiving element is configured to move the pressure element from the first position towards the second end.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/324,524, filed May 19, 2021, which is a continuation of U.S. patent application Ser. No. 15/918,762, filed Mar. 12, 2018, now U.S. Pat. No. 11,039,860, which is a continuation of U.S. patent application Ser. No. 14/490,569, filed Sep. 18, 2014, now U.S. Pat. No. 9,943,338, which claims priority to and the benefit of U.S. Provisional Patent Application Ser. No. 61/879,916, filed Sep. 19, 2013, the contents of which are hereby incorporated by reference in

BACKGROUND

Field

(1) The invention relates to a coupling assembly for coupling a rod to a bone anchoring element. The coupling assembly includes a receiving part with a recess for receiving the rod and an accommodation space for accommodating a head of the bone anchoring element, a pressure element to clamp the head, and a separate rod receiving element that is connectable to the pressure element and configured to support the rod. The pressure element has at least one spring portion that extends over a length in a circumferential direction and that is configured to engage a portion of an inner wall of the receiving part so that the pressure element can be held in a position that prevents an inserted bone anchoring element from being removed. The invention further relates to a bottom loading type polyaxial bone anchoring device with such a coupling assembly. Still further, the invention relates to a modular polyaxial bone anchoring device including the coupling assembly and at least two different rod receiving elements that can be used interchangeably with at least two associated closure mechanisms. The invention further relates to a modular stabilization device using the polyaxial bone anchoring device and at least two stabilizations rods having different diameters or one rod with at least two sections having different diameters.

Description of Related Art

(2) US 2010/0234902 A1 describes a receiving part for receiving a rod for coupling the rod to a bone anchoring element, the receiving part including a receiving part body with a channel for receiving the rod and defining an accommodation space for accommodating a head of a bone anchoring element and a pressure element at least partially provided in the accommodation space. In one embodiment the pressure element comprises two upstanding resilient fingers with outwardly directed portions at their free ends that can snap below pins extending through the wall of the receiving part in order to secure a pre-locking position of the pressure element in which the head can no longer be removed through the bottom end of the receiving part.

(3) US 2010/0160980 A1 describes a locking mechanism and a method of fixation of a bone screw and a rod to the spine. The locking mechanism includes a body, an insert, a rod seat and a set screw. The body includes a bottom portion configured to receive the bone screw and the insert, but prevents the insert and the bone screw from passing therethrough once the insert and the bone screw are engaged. The rod seat is between the rod and the insert.

SUMMARY

(4) It is an object of the invention to provide an improved coupling assembly for a bottom loading type polyaxial bone anchoring device, and such a polyaxial bone anchoring device, that allows for improved or easier handling during surgery. Furthermore, it is an object to provide a modular polyaxial bone anchoring device and a modular stabilization device that offers a greater variety of applications based on the modularity of the system.

(5) The coupling assembly includes a pressure element that has at least one spring portion extending over a length in a circumferential direction and that is compressible in a radial direction. The spring portion is configured to engage an engagement structure at an inner wall of the receiving part. In a first engagement position, the spring portion engages a first engagement structure at the inner wall of the receiving part to prevent upward movement of the pressure element in the receiving part during insertion of the head of the bone anchoring element. In a second engagement position, the spring portion engages a second engagement structure at the inner wall of the receiving part in a pre-locking position, wherein the bone anchoring element is prevented from being pulled out from a lower opening of the receiving part.

(6) In the pre-locking position of the pressure element, the head of the bone anchoring element may be held by a frictional force exerted by the pressure element onto the head. The frictional force may

be such that pivoting of the head is still possible by applying a force that overcomes the frictional force.

(7) According to an embodiment of the invention, the flexible portion of the pressure element has a circumferentially extending slit forming a slit ring or a portion of a slit ring at the bottom end of the pressure element. The slit ring can expand in a radial direction to allow for insertion of the head of the bone anchoring element. A force necessary for inserting the head into such a flexible portion of the pressure element is reduced compared to pressure elements that have, for example, a flexible portion with only longitudinal or coaxial slits. Therefore, easier assembly of the polyaxial bone anchoring device is facilitated.

(8) The coupling assembly can be assembled in situ with a bone anchoring element that has already been inserted into a bone or a vertebra.

(9) In an embodiment, the pressure element may have a recessed portion at its lower end that allows the shank of the bone anchoring element to pivot at a larger angle to one side compared to other sides of the pressure element. The pressure element further may have an indication feature configured to cooperate with an instrument to indicate the position of the recessed portion that provides the greater pivot angle. The enlarged pivot angle in one direction may be needed for special applications, for example, applications in the cervical spine.

(10) A rod receiving element may have a groove with a shape that allows support of rods or rod sections having different diameters. Therefore, a modular stabilization device is provided that includes the bone anchoring device and a variety of rods with different diameter or one or more rods that have a change in diameter over the length of the rod. This renders the polyaxial bone anchoring device suitable for many different applications that depend on different diameters of the rods to be used.

(11) In an embodiment, a first type of a rod receiving element has a first height in an axial direction that is smaller than the diameter of a rod to be supported. The first type of rod receiving element is configured to be used, for example, with a single part locking device that locks the head and the rod simultaneously.

(12) In another embodiment, the rod receiving element has a second height in an axial direction that is greater than the largest diameter of a rod to be supported. The second type of rod receiving element is configured to be used, for example, with a two part locking device that locks the head and the rod independently.

(13) A modular polyaxial bone anchoring device includes a bone anchoring element, the coupling assembly with the first type of rod receiving element and a single part locking device and the second type of rod receiving element with a two part locking device. The first type of rod receiving element with the single part locking device and the second rod receiving element with the two part locking device can be used interchangeably. The assembly of the polyaxial bone anchoring device is more easily facilitated. Therefore, a kit comprising the modular parts can be provided and part selection and assembly can be made on demand.

(14) The modular polyaxial bone anchoring device furthermore can include several bone anchoring elements that may differ in regards to the length of the shank, the anchoring features of the shank, such as different thread types, thread pitches etc., different diameters of the shank, and in regards to cannulated on non-cannulated shanks.

(15) The modularity in terms of the bone anchoring element and the type of closure element to be used, as well as the modularity in terms of the rods that can be employed, opens provides for a large variety of implants for the surgeon. In addition, the costs for stock-keeping may be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further features and advantages of the invention will become more apparent from the description of various embodiments using the accompanying drawings. In the drawings:
- (2) FIG. 1 shows a perspective exploded view of a first embodiment of a bone anchoring device;
- (3) FIG. 2 shows a perspective view of the bone anchoring device of FIG. 1 in an assembled state;
- (4) FIG. 3 shows a cross-sectional view of the bone anchoring device according to the first embodiment of FIGS. 1 and 2, the cross-section taken perpendicular to an axis of an inserted rod;
- (5) FIG. 4 shows a perspective view from above of a receiving part according to the first embodiment;
- (6) FIG. 5 shows a perspective view from the bottom of the receiving part shown in FIG. 4;
- (7) FIG. 6 shows a top view of the receiving part shown in FIGS. 4 and 5;
- (8) FIG. 7a shows a cross-sectional view of the receiving part shown in FIGS. 4 to 6 along line A-A in FIG. 6;
- (9) FIG. 7b shows an enlarged view of a detail of FIG. 7a;
- (10) FIG. 8 shows a perspective view from above of a pressure element according to a first embodiment;
- (11) FIG. 9 shows a perspective view from the bottom of the pressure element shown in FIG. 8;
- (12) FIG. 10 shows a side view of the pressure element shown in FIGS. 8 and 9;
- (13) FIG. 11 shows a top view of the pressure element shown in FIGS. 8 to 10;
- (14) FIG. 12 shows a cross-sectional view of the pressure element of FIGS. 8 to 11 along line B-B in FIG. 11;
- (15) FIG. 13 shows a perspective view from above of a rod receiving element according to a first embodiment;
- (16) FIG. 14 shows a perspective view from the bottom of the rod receiving element of FIG. 13;
- (17) FIG. 15 shows a top view of the rod receiving element shown in FIGS. 13 and 14;
- (18) FIG. 16 shows a cross-sectional view of the rod receiving element of FIGS. 13 to 15 along line D-D in FIG. 15;
- (19) FIG. 17 shows a cross-sectional view of a first step of assembling the coupling assembly according to the first embodiment;
- (20) FIG. 18a shows a cross-sectional view of a second step of assembling the coupling assembly according to the first embodiment;
- (21) FIG. 18b shows an enlarged cross-sectional view of a detail of FIG. 18a;
- (22) FIG. 19 shows a cross-sectional view of a first step of assembling the polyaxial bone anchoring device (e.g., with a bone anchoring element) according to the first embodiment;
- (23) FIG. 20 shows a cross-sectional view of a second step of assembling the polyaxial bone anchoring device according to the first embodiment;
- (24) FIG. 21a shows a cross-sectional view of a third step of assembling the polyaxial bone anchoring device according to the first embodiment;
- (25) FIG. 21b shows an enlarged view of a detail of FIG. 21a;
- (26) FIG. 22 shows a perspective view of a pressure element of a coupling assembly according to a second embodiment;
- (27) FIG. 23 shows a side view of the pressure element shown in FIG. 22;
- (28) FIG. 24 shows a perspective view from above of a rod receiving element of a coupling assembly according to a second embodiment;
- (29) FIG. 25 shows a perspective view from the bottom of the rod receiving element shown in FIG. 24;
- (30) FIG. 26 shows a top view of the rod receiving element shown in FIGS. 24 and 25;
- (31) FIG. 27 shows a cross-sectional view of the rod receiving element shown in FIGS. 24 to 26 along line G-G in FIG. 26;

(32) FIG. 28 shows a perspective schematic view of a modular polyaxial bone anchoring device;
(33) FIG. 29a shows a cross-sectional view of the polyaxial bone anchoring device according to the first embodiment with an inserted rod having a first diameter and a single part locking device;
(34) FIG. 29b shows a cross-sectional view of the polyaxial bone anchoring device according to the first embodiment with an inserted rod having a second diameter and a single part locking device;
(35) FIG. 30a shows a cross-sectional view of a polyaxial bone anchoring device according to the second embodiment with an inserted rod having a first diameter and a two part locking device; and
(36) FIG. 30b shows a cross-sectional view of the polyaxial bone anchoring device according to the second embodiment with an inserted rod having a second diameter and a two part locking device.

DETAILED DESCRIPTION

(37) Referring to FIGS. 1 and 2, a bone anchoring device according to a first embodiment
(38) includes a bone anchoring element **1** in the form of a bone screw having a shank **2** that is at least partially provided with a bone thread, and a head **3**. The head **3** has a spherical segment-shaped surface portion including a greatest outer diameter E of the sphere and a flat free end with a recess **4** for engagement with a screwing-in tool.
(39) The bone anchoring device further includes a coupling assembly **5** for receiving a stabilization rod **100** and for coupling the stabilization rod **100** to the bone anchoring element **1**. The coupling assembly **5** includes a receiving part **6** for receiving the head **3** of the bone anchoring element **1** and for receiving the rod **100**, and a pressure element **7** configured to be arranged in the receiving part **6**. The pressure element **7** is utilized for locking the head **3** in the receiving part **6**. The coupling assembly further includes a rod receiving element **8** that is connectable to the pressure element **7** and that serves for providing support for an inserted rod **100**. A pin **9** may be employed for securing a rotational position of the rod receiving element **8**, so that a support surface for the rod **100** is aligned with a recess of the receiving part **6** through which the rod **100** extends as further described below.
(40) Further, a locking element **10** in the form of an inner screw is provided for securing the rod **100** in the receiving part **6** and for exerting a force via the rod **100** onto the rod receiving element **8** and the pressure element **7** to lock the head **3** in the receiving part **6**. The locking element **10** in this embodiment is a single part locking element that is configured to lock the head **3** of the bone anchoring element **1** and the rod **100** simultaneously.
(41) The receiving part **6** will now be explained with reference to FIGS. 1 to 7b. The receiving part **6** has a first end **6a** that is a top end and an opposite second end **6b** that forms a bottom end, and a central axis of symmetry C passing through the first end **6a** and the second end **6b**. A bore **61** is provided that is coaxial with the central axis C. In a first region adjacent to the first end **6a** the receiving part **6** has a substantially U-shaped recess **62** with a bottom directed towards the second end **6b** and two free lateral legs **62a**, **62b** extending towards the first end **6a**. On the legs **62a**, **62b**, and internal thread **63** is provided that cooperates with the locking element **10**. The channel formed by the U-shaped recess **62** is sized so as to receive the rod **100** therein for connecting at least two bone anchoring devices. In the region of the legs **62a**, **62b** to substantially a height in an axial direction defined by the bottom of the U-shaped recess **62**, the bore **61** has a first diameter. In a region below the legs **62a**, **62b**, the bore **61** has a widened portion **61a** with a diameter greater than the first diameter of the bore **61**. Between the second end **6b** and the widened portion **61a**, the bore **61** has a narrowing portion **61b** that tapers and narrows towards the second end **6b** with a cone angle. An opening **64** is provided at the second end **6b**, the diameter of the opening **64** being larger than the largest diameter E of the head **3** to allow the insertion of the head **3** from the second end **6b** of the receiving part **6**. The widened portion **61a** and the narrowing portion **61b** define an accommodation space for the head **3** of the bone anchoring element **1**.
(42) Between the portion of the bore **61** with the first diameter that is located adjacent to the first end **6a** and the widened portion **61a** of the bore **61**, there is a recess **65** that extends circumferentially at the inner wall of the receiving part **6** and that has a substantially conical shape

widening in a direction from the second end **6b** towards the first end **6a**. As can be seen in detail in FIG. **7b**, the size of the recess **65** is such that, at the end facing the second end **6b** of the receiving part **6**, the recess **65** has an inner diameter corresponding substantially to the first diameter of the portion of the bore **61** nearer to the first end **6a**, and therefore forms an annular inwardly protruding edge **66** at the upper end of the widened portion **61a**. The protruding edge **66** defines a first stop. An upper or opposite end or region of the recess **65** has a greater diameter than the portion of the bore **61** nearer to the first end **6a**, and forms a second edge **67** that defines a second stop. A cone angle of the recess **65** is such that a spring portion provided at the pressure element **7** can be held in the recess **65** in a pre-stressed manner, as further described below.

(43) At at least one of the legs **62a**, **62b**, a transverse bore **68** is provided that extends through the leg, for example through the leg **62b**, in a direction substantially perpendicular to the central axis C for receiving the pin **9**. The pin **9** has a length such that once the pin **9** is inserted into the transverse bore **68**, the pin **9** extends a short distance into the bore **61** to provide a stop for securing a rotational position of the rod receiving element **8**, as further described below. The pin **9** may be flush with an outer surface of the receiving part **6** when inserted.

(44) Referring to FIGS. **8** to **12**, the pressure element **7** has a first end **7a** and a second end **7b**. The second end **7b** of the pressure element **7** is configured to be closer to the second end **6b** of the receiving part **6** than the first end **7a** of the pressure element **7** is to the second end **6b** of the receiving part **6** when the pressure element **7** is arranged in the receiving part **6**. The pressure element **7** is a substantially cap-like part that has a hollow interior chamber **71** with an opening **72** at the second end **7b**, wherein the hollow interior chamber **71** is configured and sized to accommodate and hold the head **3** of the bone anchoring element **1** therein. An opening **73** near the first end **7a** that communicates with the hollow interior chamber **71** allows access to the recess **4** of an inserted head **3** with a tool from the first end **7a** of the pressure element **7**. The lower opening **72** is sized to allow the insertion of the head **3** from the second end **7b**. A lower portion **71a** of the pressure element **7** adjacent to the second end **7b** tapers or narrows towards the second end **7b** with an outer surface, which in this embodiment matches or corresponds substantially to the shape of the narrowing portion **61b** of the receiving part **6**.

(45) At a distance from the second end **7b**, a circumferentially extending slit **74** is provided. The slit **74** extends around the central axis C of the pressure element **7** along a plane substantially perpendicular to the central axis C. Further, the slit **74** extends around more than 180°, and preferably more than 270°, and further preferably more than 340° around the central axis C. By means of the slit **74**, a ring-shaped portion at the second end **7b** is provided that is integrally connected to the rest of the pressure element **7** by a wall portion forming a connecting strip **75**. The connecting strip **75** has a length in the circumferential direction such that it provides a stable connection of the ring-shaped portion to the rest of the pressure element **7**. At one end of the circumferentially extending slit **74**, there is a substantially vertical slit **76** that extends from the second end **7b** fully through the ring-shaped portion into the circumferentially extending slit **74**. By means of this, the ring-shaped portion is cut through or split in a circumferential direction and forms a slit ring **77** that can be expanded and compressed in a radial direction. The outer surface of the slit ring **77** forms the narrowing outer surface of the lower portion **71a** that narrows towards the second end **7b** of the pressure element **7**. A position and size of the slit ring **77** is such that when the head **3** of the bone anchoring element **1** is inserted through the opening **72** into the hollow interior chamber **71**, the slit ring **77** expands so that the width of the vertical slit **76** becomes larger, and when the head **3** has been fully inserted into the hollow interior chamber **71**, the slit ring **77** encompasses the head **3** at or below a position of the largest diameter E of the head **3** in a direction towards the shank **2**. An inner surface of the pressure element **7** in the region of the slit ring **77** may be spherical segment-shaped, with a size that is adapted or corresponds to the size of the head **3**.

(46) A portion **71b** of the pressure element **7** adjacent to the slit ring **77** is substantially cylindrical with an outer diameter that is smaller than an inner diameter of the widened portion **61a** of the

receiving part **6**. A corresponding inner surface of the portion **71b** is substantially cylindrical with an inner diameter corresponding to or slightly larger than the largest outer diameter **E** of the head **3**. (47) The pressure element **7** further has an intermediate portion **71c** that continues from or is connected to the cylindrical portion **71b**. The intermediate portion **71c** has a conical outer surface that widens towards the first end **7a** of the pressure element **7**. An outermost edge **78** of the intermediate portion **71c** that faces the first end **7a** protrudes outward from the rest of the pressure element **7**. The cone angle of the intermediate portion **71c** corresponds substantially to the cone angle of the conical recess **65** of the receiving part **6**. An inner surface of the intermediate portion **71c** is cylindrical and has an inner diameter corresponding to the inner diameter of the cylindrical portion **71b**.

(48) Next, an upper cylindrical portion **71d** extends from the intermediate portion **71c** towards the first end **7a** of the pressure element **7**. The upper cylindrical portion **71d** has an outer diameter that may be substantially the same as the outer diameter of the lower cylindrical portion **71b** and that is in particular slightly smaller than the inner diameter of the bore **61**. The upper cylindrical portion **71d** defines the coaxial bore **73** and a spherical segment-shaped section **710** that may make up part of the hollow interior chamber **71** and has a radius adapted to the spherical segment-shaped head **3** of the bone anchoring element **1**. Hence, when the head **3** of the bone anchoring element **1** is inserted into the hollow interior chamber **71** and tilted, as can be seen, for example, in FIG. **3**, the spherical segment-shaped section **710** contacts the surface of the head **3**.

(49) In the intermediate portion **71c**, two spring portions **79a**, **79b** are provided. The spring portions **79a**, **79b** are ring segment-shaped and extend around the central axis **C** around an angle α of approximately 90° , as can be seen in particular in FIG. **11**. However, the ring segment-shaped spring portions **79a**, **79b** can extend around a smaller or a larger angle around the central axis **C**, depending on the desired flexibility to be achieved.

(50) The spring portions **79a**, **79b** are formed by slits that extend substantially coaxially to the central axis **C** and slits that extend substantially in a circumferential direction around the pressure element **7**. Referring in particular to FIG. **10**, the spring portion **79a** is formed by the following slits. A first vertical slit **791a** extends from the lower cylindrical portion **71b** of the pressure element **7** through the intermediate portion **71c** and to approximately the upper cylindrical portion **71d** in a substantially vertical or coaxial direction with respect to the central axis **C**. A second vertical slit **792a** is spaced apart from the first vertical slit **791a** in a circumferential direction and extends from the lower cylindrical portion **71b** of the pressure element **7** to the intermediate portion **71c**. A first or upper horizontal slit **793a** extends from the first vertical slit **791a** in a circumferential direction a distance around the pressure element **7** that defines the angle α of the spring portion around the central axis **C**. A second or lower horizontal slit **794a** extends from the upper end of the second vertical slit **792a** in a circumferential direction a distance around the pressure element **7** that is slightly less than the angle α around the central axis **C**. A third vertical slit **795a** connects the ends of the first horizontal slit **793a** and the second horizontal slit **794a** on a side opposite the first and second vertical slits **791a**, **792a**. Widths of the vertical slits **791a**, **792a**, **795a** in a circumferential direction may be greater than widths of the horizontal slits **793a**, **794a** in an axial direction. Further, the width of the upper horizontal slit **793a** may be the same as the width of the lower horizontal slit **794a**.

(51) By the first and second vertical slits **791a**, **792a**, a strip is formed that connects the ring segment-shaped spring portion **79a** with the rest of the pressure element **7**. By the third vertical slit **795a**, the ring segment-shaped spring portion **79a** is provided with a free end in the circumferential direction.

(52) The second ring segment-shaped spring portion **79b** is formed identically to the first spring portion **79a** and has a first vertical slit **791b**, a second vertical slit **792b**, and a third vertical slit **795b**, as well as a first upper horizontal slit **793b** and a second lower horizontal slit **794b**. The ring segment-shaped spring portions **79a**, **79b** are arranged mirror-symmetrical with respect to a mirror

plane that contains the central axis C and extends between the first vertical slit **791a**, **791b** of each spring portion, as depicted in particular in FIG. 12. Hence, the free ends of the ring segment-shaped spring portions **79a**, **79b** point towards each other in the circumferential direction.

(53) The spring portions **79a**, **79b** are compressible and expandable in a radial direction. When the pressure element **7** is inserted into the bore **61** of the receiving part **6**, the spring portions **79a**, **79b** are slightly compressed so that the pressure element **7** can move downward in the bore **61** (i.e., towards the second end **6b**). Once the spring portions **79a**, **79b** have passed the portion of the bore **61** having the first diameter, they are configured to resiliently snap into the conical recess **65**.

(54) The pressure element **7** further includes adjacent to the first end **7a**, a plurality of upstanding ring segment-shaped projections **71e** that are spaced apart in a circumferential direction and that are slightly resilient. An outer surface of the projections **71e** may be slightly tapered so as to provide for a safe connection with the rod receiving element **8**, as further described below. An inner surface of the projections **71e** may be cylindrical. The upstanding projections **71e** are spaced apart from or recessed from an outer surface of the upper cylindrical portion **71d**, and surround the coaxial bore **73**.

(55) Adjacent to the second end **7b**, there is a recessed portion **796** at the lower edge of the pressure element **7** that is formed by the lower edge of the slit ring **77**. The recessed portion **796** has a shape and size such that the shank **2** of the bone anchoring element **1** can abut at a larger pivot angle against the lower edge of the pressure element **7** when bone anchoring element **1** pivots in the direction of the recessed portion **796**, as compared to other directions. Therefore, the recessed portion **796** provides for an enlarged pivot angle in one direction. As can be seen in FIG. 10, the recessed portion **796** may be aligned approximately at a middle of one of the spring portions **79a**, **79b**. A second recessed portion **797** is provided at a corresponding circumferential position in the inner wall of the upper cylindrical portion **71d**. The second recessed portion **797** serves for indicating the position of the first recessed portion **796**, for example, when the pressure element **7** is engaged with a tool (not shown) that includes an engagement portion to engage the second recessed portion **797**. Such a tool could have a visual indication on its handle, for example, that shows the user the position of the second recessed portion **797**, and as a consequence thereof, the position of the first recessed portion **796**. Hence, the position of the first recessed portion **796** that provides for the enlarged pivot angle can be identified even if the polyaxial bone anchoring device has already been implanted into a bone or a vertebra. The tool can also be used to engage the second recessed portion **797** in a form fit manner, and to rotate the pressure element **7** to adjust the orientation or direction of the first recessed portion **796**.

(56) With reference to FIGS. 3 and 13 to 16, the rod receiving element **8** will be described. The rod receiving element **8** is a substantially hollow cylindrical part having a first end **8a** and an opposite second end **8b**. A groove **81** having a substantially V-shaped cross section with a rounded bottom **81a** extends from the first end **8a** in the direction of the second end **8b**, as can be seen in particular in FIGS. 13 and 16. The groove **81** defines a longitudinal direction that is parallel to the longitudinal axis of the rod **100** when the rod **100** is inserted. The groove **81** provides a support surface for supporting the rod **100**. Referring to FIG. 3, the rod **100** has a circular cross section and is supported in the groove **81** along two contact lines **P1**, **P2**, that extend in a direction parallel to the rod axis. Depending on the diameter of the rod section that is supported by the groove **81**, the contact lines **P1**, **P2** are located more towards the bottom **81a** of the groove **81** for a rod with a smaller diameter, or more towards the first end **8a** of the groove **81** for a rod with a larger diameter, when compared to the rod **100** depicted in FIG. 3. Hence, the groove **81** is configured to support different rods having different diameters or different rod sections of a single rod that have different diameters.

(57) Adjacent to the second end **8b**, two opposite annular segment-shaped protrusions **82a**, **82b** are provided that extend towards the central axis C in a radial direction and along a certain length in the circumferential direction. A height of the annular segment-shaped protrusions **82a**, **82b** in the

axial direction measured from the second end **8b** corresponds to a height of the upstanding projections **71e** of the pressure element **7**. Furthermore, an inner surface of the annular segment-shaped protrusions **82a**, **82b** is slightly tapered and extends closer to the central axis C towards the second end **8b**, and is configured to cooperate with the tapered outer surface of the upstanding projections **71e** of the pressure element **7**. Once the annular segment-shaped protrusions **82a**, **82b** engage the outer wall of the upstanding projections **71e** of the pressure element **7**, the upstanding projections **71e** are slightly compressed and abut against the annular segment-shaped protrusions **82a**, **82b** so that the pressure element **7** is connected to the rod receiving element **8** and held by a frictional force. The location of the center of the annular segment-shaped protrusions **82a**, **82b** in a circumferential direction is at substantially 90° with respect to the bottoms **81a** of the groove **81**.

(58) In the outer wall of the rod receiving element **8** on one side of the groove **81**, an elongate recess **83** that extends substantially parallel to the central axis C is provided. The elongate recess **83** has a closed end **84** towards the first end **8a** that serves as an abutment for the pin **9**. By the elongate recess **83**, a securing element is provided that also secures a correct or desired rotational position of the rod receiving element **8**. The elongate recess **83** is open towards the second end **8b**.

(59) In the described embodiment, depth of the groove **81** is smaller than a largest diameter of a rod that can be supported by the rod receiving element **8**. In other words, the first end **8a** will be located below a top surface of a rod with a largest diameter that can be supported by the rod receiving element **8**.

(60) An outer diameter of the rod receiving element **8** is only slightly smaller than an inner diameter of the bore **61** of the receiving part and is preferably flush with an outer diameter of the upper cylindrical portion **71d** of the pressure element **7** when the pressure element **7** and the rod receiving element **8** are assembled together as shown in FIG. 3.

(61) The bone anchoring device, as a whole or partially, is made of a bio-compatible material, such as a bio-compatible metal or a metal alloy, for example titanium, stainless steel, of a nickel titanium alloy, for example, nitinol, or of bio-compatible plastic materials, such as, for example, polyetheretherketone (PEEK).

(62) Referring to FIG. 17, the coupling assembly **5** is assembled by pre-assembling the pressure element **7** and the rod receiving element **8**, such that the upstanding projections **71e** are inserted and held between the annular segment-shaped protrusions **82a**, **82b** of the rod receiving element **8**. The orientation of the pressure element **7** relative to the groove **81** of the rod receiving element **8** is such that the center of the spring portions **79a**, **79b** may initially be located in a circumferential direction at an angle of substantially 90° measured from the longitudinal axis of the groove **81**. The pre-assembled pressure element **7** with the rod receiving element **8** is inserted from the first end **6a** of the receiving part, with the second end **7b** of the pressure element **7** facing in the direction of the second end **6b** of the receiving part **6**. During insertion, the spring portions **79a**, **79b** are compressed in a radial direction because the inner diameter of the portion of the bore **61** with the first diameter in the receiving part **6** has a smaller diameter than the outermost edge **78** of the spring portions **79a**, **79b**.

(63) Referring to FIGS. 18a and 18b, as soon as the pressure element **7** is in a position in which the outermost edge **78** of the spring portions **79a**, **79b** has reached the conical recess **65** in the receiving part **6**, the spring portions **79a**, **79b** snap behind the protrusion **67** that forms the upper edge of the conical recess **65**. Thereby, the spring portions **79a**, **79b** are configured to radially expand into the conical recess **65**. Thereafter, the pin **9** is inserted into the transverse bore **68** until a front face of the pin **9** extends into the elongate recess **83** provided at the rod receiving element **8**. The pin **9** serves for securing the pressure element **7** and the rod receiving element **8** against inadvertent rotation, so that the U-shaped recess **62** of the receiving part and the groove **81** of the rod receiving element **8** remain aligned. In this condition, the coupling assembly **5** is pre-assembled and can be used for coupling to a bone anchoring element **1** and to a rod **100**. In the pre-assembled condition, it is still possible to rotate the pressure element **7** with respect to the rod receiving

element **8**, and therefore also with respect to the receiving part **6**, so that the orientation of the first recessed portion **796** that defines the direction having the enlarged pivot angle can be adjusted.

(64) Referring further to FIGS. **19** and **20**, the use of the coupling assembly **5** together with a bone anchoring element **1** will be explained. As depicted in FIG. **19**, first, a suitable bone anchoring element **1** is selected. The bone anchoring element **1** may be connected to the coupling assembly **5** first, and thereafter inserted into a bone part or a vertebra. Alternatively, the bone anchoring element **1** can be placed first into the implantation site in a patient's body without the coupling assembly **5** being connected thereto. As shown in FIG. **19**, the head **3** enters the receiving part **6** through the lower opening **64** and enters the hollow interior chamber **71** of the pressure element **7** through the open second end **7b** of the pressure element **7**. When the head touches the slit ring **77** of the pressure element **7**, the pressure element **7** can not move upward towards the first end **6a** of the receiving part **6** because the upper surfaces of the spring portions **79a**, **79b** abut against the upper edge **67** of the conical recess **65** that forms the second stop.

(65) As shown in FIG. **20**, further insertion of the head **3** upwards into the hollow interior chamber **71** expands the slit ring **77** within the widened portion **61a** of the bore **61** of the receiving part **6**. The head **3** can then be completely inserted into the pressure element **7**. Because the slit ring **77** does not expand at the connection strip **75** the insertion of the head **3** may not be precisely coaxial with the central axis C, but instead may be slightly out of or misaligned with the central axis C. By the further insertion of the head **3**, the slit ring **77** may be expanded to a maximum extent to allow the head **3** to enter the upper portion of the hollow interior chamber **71** until the head **3** rests in the spherical segment-shaped portion of the chamber **71** that is adjacent to the second end **7b**. Here, the slit ring **77** can elastically contract around the head **3** as shown in FIG. **20**.

(66) Referring to FIGS. **21a** and **21b**, pulling the receiving part **6** upwards as illustrated, and/or pressing down the pressure element **7** with an instrument (not shown) presses the slit ring **77** into the narrowing portion **61b** of the receiving part **6**. The conical shape of the recess **65** at the inner wall of the receiving part **6** provides an inclined surface that allows the spring portions **79a**, **79b** to slide along the inner wall when the pressure element **7** is further moved downwards towards second end **6b**, thereby gradually compressing the spring portions **79a**, **79b** again until the upper outer edges **78** of the spring portions **79a**, **79b** snap behind the second inner protrusions **66** in the receiving part **6**. In this position, the pressure element **7** is again prevented from moving upwards towards the first end **6a** of the receiving part **6** by the first stop provided by the inner protruding edge **66**. The head **3** is already clamped by the slit ring **77**. Because the slit ring **77** is located between the head **3** and the narrowing portion **61b** of the receiving part **6**, the slit ring **77** is prevented from expanding, and so the head **3** is prevented from falling out or being pushed out through the lower opening **64**. This is the pre-locking condition or position.

(67) In clinical use, usually at least two bone anchoring devices are inserted into the bone and the respective receiving parts **6** are aligned. The heads **3** of the bone anchoring elements **1** are held in the respective pressure elements **7** by a frictional force. Hence, the receiving parts **6** can be easily aligned manually, and their angular positions are maintained by the frictional force between the heads **3** and the pressure elements **7**.

(68) Finally, referring to FIG. **3**, the rod **100** is inserted into the receiving part **6**. The rod **100** rests on the upper surface of the groove **81** at substantially two longitudinal contact areas P1, P2. Then, the locking element **10** is screwed between the legs **62a**, **62b** of the receiving part **6**. Tightening of the locking element **10** advances the locking element **10** towards the rod **100** until it contacts an upper surface of the rod **100**. The downward force applied by the locking element **10** is transferred from the rod **100** to the pressure element **7**. Finally, the head **3** and the rod **100** are locked relative to the coupling assembly **5**. The head **3** can be locked in a specific angular position with respect to the receiving part **6**.

(69) The bone anchoring element **1** can be pivoted in the direction of the recessed portion **796** at the second end **7b** of the pressure element **7** to a greater angle relative to the receiving part **6** than in

an opposite or other directions.

(70) Referring to FIGS. 22 and 23, a second embodiment of the pressure element will be described. The pressure element 7' differs only in that the spring portions 79a', 79b' do not have a free end. Instead, each spring portion 79a', 79b', is attached at the respective ends of the ring segment-shaped portions via corresponding axially extending strips that are formed by two vertical slits 798a, 799a, 798b, 799b, respectively, wherein slits 798a, 798b extend to the first horizontal slit 793a, 793b, and slits 799a, 799b extend to the second horizontal slit 794a, 794b, respectively.

(71) A second embodiment of the rod receiving element will be described with reference to FIGS. 24 to 27. Parts and portions that are identical or similar to those of the first embodiment have the same reference numerals, and the descriptions thereof will not be repeated. The rod receiving element 8' includes two opposite upstanding legs 86a, 86b that extend upward from the rod supporting groove 81. The upstanding legs 86a, 86b are provided by a substantially rectangular recess 86 that is cut into the hollow cylindrical rod receiving element 8' starting from the first end 8a towards the second end 8b. The upstanding legs 86a, 86b have a height such that they extend above a top surface of an inserted rod 100 that rests in the rod supporting groove 81. This renders the rod receiving element 8' suitable for use with a two-part locking device 10' as depicted in FIG. 28.

(72) The two-part locking device 10' includes an outer locking element 10a and an inner locking element 10b. The outer locking element 10a cooperates with the internal thread 63 of the receiving part 6 and is configured to abut against the first end 8a of the rod receiving element 8' according to the second embodiment. The inner locking element 10b can be screwed into a threaded hole of the outer locking element 10a and is configured to cooperate with the rod but not with the inserted rod receiving element 8'. When the two-part locking element 10' is used, the head 3 of the bone anchoring element 1 can be locked by exerting pressure with the outer locking element 10a only onto the rod receiving element 8' according to the second embodiment, and via the pressure element 7, 7' onto the head 3, while the rod can be locked independently by exerting pressure with the inner locking element 10b onto its surface, independently from the locking of the head 3.

(73) FIG. 28 depicts a modular device for stabilizing bones or vertebrae, and includes a modular polyaxial bone anchoring device and different kinds of stabilization rods 100, 100' that have different rod diameters. The modular polyaxial bone anchoring device includes the receiving part 6 and an assembly kit including the rod receiving element 8 according to the first embodiment and the rod receiving element 8' according to the second embodiment that can be used interchangeably together with the pressure element 7, 7'. Corresponding to the first type of coupling assembly utilizing the rod receiving element 8, the single part locking device 10 can be used as part of the modular device, and corresponding to the second type of coupling assembly utilizing the rod receiving element 8', the two-part locking device 10' can be used as part of the modular device. In other embodiments, various other types of locking devices that are configured to cooperate with either the rod receiving element 8 or the rod receiving element 8' may also be employed.

(74) A modular polyaxial bone anchoring device further includes at least one bone anchoring element 1, preferably a plurality of bone anchoring elements that may differ with respect to the lengths of the shanks, anchoring features of the shanks, such as different thread types, thread pitches, different diameters of the shanks, cannulated or non-cannulated shanks, or with respect to various other features. Because the polyaxial bone anchoring device is a bottom loading type polyaxial bone anchoring device, the assembly of the polyaxial bone anchoring device is easily made, for example, by the surgeon or any personnel assisting the surgeon, so that a suitable bone anchoring device can be provided on demand during or before surgery.

(75) The modular stabilization device may further include at least two rods 100, 101 having different diameters and/or at least one rod having sections with different diameters.

(76) FIGS. 29a to 30b show various combinations of the elements of the modular system depicted in FIG. 28. In FIG. 29a, the first type of coupling assembly using the first type of rod receiving

element **8** is used together with a rod **100** of a larger diameter and a single part locking device **10**. (77) The locking device **10** exerts pressure onto the rod **100** and locks the rod and the head **3** simultaneously. In FIG. **29b**, the first type of coupling assembly as shown in FIG. **29a** is used together with a rod **101** having a smaller diameter than the rod **100** shown in FIG. **29a**. The single part locking device **10** exerts pressure onto the rod **101** with a smaller diameter and simultaneously locks the head **3** and the rod **101**. The rod **101** with the smaller diameter is safely clamped in the same manner as the rod **100** with the larger diameter.

(78) In FIG. **30a**, the second type of coupling assembly using the second type or rod receiving element **8'** is used together with the rod **100** having a larger diameter and with a two-part locking device **10'**. The outer locking element **10a** acts on the first end **8a** of the rod receiving element **8'**, which in turn transfers the force onto the pressure element **7** and locks the head **3**. The rod **100** can still be moved in an axial direction, and can be locked independently by the inner locking element **10b** that exerts pressure onto the rod **100** but not onto the head **3**. FIG. **30b** shows the same situation as in FIG. **30a**, with the only difference being that the rod **101** having a smaller diameter than the rod **100** shown in FIG. **30a** is used. The inner locking element **10b** has to be screwed deeper into the outer locking element **10a** in order to clamp the rod **101**.

(79) Various other modifications of the embodiments described above may also be contemplated. For example, the receiving part is not limited to the exact shape as shown. The recess **62** of the receiving part does not have to have an exact U-shape. The bore **61** can also have several sections with different widths, as long as the enlarged portion **61a** that provides space for the expansion of the pressure element is provided. The narrowing portion at the bottom of the receiving part is shown to be tapered, but can also be rounded, for example. Also, the external surface of the lower portion at the bottom end of the pressure elements **7**, **7'** can be rounded. Combinations of the surfaces of the receiving part and the pressure element that cooperate to clamp the head can, for example, be tapered and tapered, tapered and rounded, rounded and tapered, or rounded and rounded, among other configurations.

(80) The embodiments have been described with one single ring segment-shaped section on the pressure element that clamps the head. However, there may be more than one ring segment-shaped section to clamp the head in other embodiments.

(81) Meanwhile, the horizontal and vertical slits in the pressure element need not to be exactly horizontal or exactly vertical, and instead, they may have inclinations or shapes that differ from a straight or a circular shape, in order to achieve different elastic properties.

(82) In other embodiments, the interior hollow chamber of the pressure element and/or the head of the bone anchoring element can have different shapes that restrict the pivoting of the bone anchoring element relative to the receiving part or the pressure element to one single plane, so that the pivot connection is not polyaxial, but is instead monoplanar.

(83) It is also possible to provide more than one, for example two or three, recessed portions at the second end of the pressure element, for having more than one direction that allows for an enlarged pivot angle. The recessed portion at the second end of the pressure element can also be omitted, so that the pivot angle is the same in all directions.

(84) In addition, all kinds of rods can be used. While rods with a smooth surface are shown, roughened rods or rods having other structures may also be used. The rods may also be rods made of a flexible material, or may have flexibility through other means.

(85) While a number of different embodiments are disclosed herein, it is appreciated that different components from the different embodiments can be mixed and matched to produce a variety of still other different embodiments.

(86) While the present invention has been described in connection with certain exemplary embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but is instead intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, and equivalents thereof.

Claims

1. A bone anchoring device for coupling a rod to bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first and second ends, a recess at the first end for receiving the rod, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; a rod receiving element separate from and movable in the receiving part, the rod receiving element comprising a rod engagement surface configured to extend into the recess of the receiving part to engage the rod; a seat portion separate from the receiving part and the rod receiving element, wherein the seat portion is positionable in the accommodation space and comprises an upwardly facing surface configured to directly engage and exert an upward pressure directed towards the first end of the receiving part on the head; and a spring portion separate from the receiving part and the rod receiving element, the spring portion comprising an engagement surface, wherein at least part of the spring portion is configured to be positioned at an axial position that is lower than the entire rod receiving element in the receiving part; wherein when the bone anchoring device is at an insertion position, the seat portion is expandable while a position of the entire spring portion remains constant to facilitate passing of the head upwardly past the seat portion when the head is inserted through the opening of the receiving part; and wherein the bone anchoring device is adjustable from the insertion position to a pre-locking position where the seat portion is restricted from expanding to prevent the head from being removed through the opening of the receiving part, while the engagement surface of the spring portion engages a corresponding surface of the receiving part to restrict adjustment of the bone anchoring device back towards the insertion position.
2. The bone anchoring device of claim 1, wherein the seat portion is ring segment-shaped.
3. The bone anchoring device of claim 1, wherein the spring portion is ring segment-shaped.
4. The bone anchoring device of claim 1, wherein the entire spring portion is positionable at the axial position that is lower than the entire rod receiving element in the receiving part.
5. The bone anchoring device of claim 1, wherein the seat portion and the spring portion are formed as part of a monolithic pressure element.
6. The bone anchoring device of claim 1, wherein the spring portion is expandable radially relative to the central axis in the receiving part.
7. The bone anchoring device of claim 6, wherein at least part of the spring portion is configured to expand radially farther from the central axis than the entire rod receiving element.
8. The bone anchoring device of claim 1, wherein the engagement surface of the spring portion is configured to engage a second surface of the receiving part that is different from the corresponding surface of the receiving part when the bone anchoring device is at the insertion position to restrict movement of the spring portion towards the first end of the receiving part.
9. The bone anchoring device of claim 1, wherein the spring portion and the seat portion are configured to be spaced apart axially from one another.
10. A bone anchoring device for coupling a rod to bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first and second ends, a recess at the first end for receiving the rod, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; a rod receiving element separate from and movable in the receiving part, the rod receiving element comprising a rod engagement surface configured to extend into the recess of the receiving part to engage the rod; and a pre-locking element separate from the receiving part and the rod receiving element, the pre-locking

element comprising an engagement surface, wherein the pre-locking element is positionable below the rod receiving element in the receiving part, and wherein at least part of the pre-locking element is configured to be positioned at an axial position that is higher than a bottom of the rod receiving element in the receiving part; wherein when the rod receiving element and the pre-locking element are in the receiving part, the pre-locking element is configured to assume an insertion position where the head of the bone anchoring element is insertable through the opening of the receiving part into the accommodation space and where the engagement surface of the pre-locking element engages a first abutment of the receiving part to restrict movement of the pre-locking element towards the first end of the receiving part; and wherein the pre-locking element is movable axially downwardly from the insertion position to a pre-locking position where the head is prevented from being removed through the opening of the receiving part and where the engagement surface of the pre-locking element engages a second abutment of the receiving part to restrict movement of the pre-locking element back towards the insertion position.

11. The bone anchoring device of claim 10, wherein the pre-locking element is configured to directly engage the head to clamp the head in the receiving part.

12. The bone anchoring device of claim 10, wherein the part of the pre-locking element configured to be positioned at an axial position that is higher than a bottom of the rod receiving element is configured to extend into the rod receiving element.

13. The bone anchoring device of claim 10, wherein a portion of the pre-locking element where the engagement surface is formed is ring segment-shaped.

14. The bone anchoring device of claim 10, wherein a portion of the pre-locking element where the engagement surface is formed is expandable radially relative to the central axis in the receiving part.

15. The bone anchoring device of claim 10, wherein the rod receiving element and the pre-locking element are configured to directly engage one another in the receiving part.

16. The bone anchoring device of claim 10, wherein the pre-locking element is movable axially relative to the receiving part.

17. A bone anchoring device for coupling a rod to bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first and second ends, a recess at the first end for receiving the rod, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; a rod receiving element separate from and movable in the receiving part, the rod receiving element comprising a rod engagement surface configured to extend into the recess of the receiving part to engage the rod; a seat portion separate from the receiving part and the rod receiving element, wherein the seat portion is positionable in the accommodation space and comprises an upwardly facing surface configured to directly engage and exert an upward pressure directed towards the first end of the receiving part on the head; and a spring portion separate from the receiving part and the rod receiving element, wherein when the seat portion and the spring portion are in the receiving part, a part of the spring portion that is ring segment-shaped extends circumferentially around the central axis with an upwardly facing surface and a downwardly facing surface each separable from other parts of the bone anchoring device, and the ring segment-shaped part of the spring portion is configured to be spaced apart axially from the seat portion; wherein when the bone anchoring device is at an insertion position, the seat portion is expandable to facilitate passing of the head upwardly past the seat portion when the head is inserted through the opening of the receiving part; and wherein the bone anchoring device is adjustable from the insertion position to a pre-locking position where the seat portion is restricted from expanding to prevent the head from being removed through the opening of the receiving part, while the spring portion is moved axially relative to the receiving part to a position where the spring portion restricts adjustment of the bone anchoring device back

towards the insertion position.

18. The bone anchoring device of claim 17, wherein the seat portion and the spring portion are formed as part of a monolithic pressure element.

19. The bone anchoring device of claim 17, wherein the spring portion is expandable radially relative to the central axis in the receiving part.

20. The bone anchoring device of claim 17, wherein the spring portion and the seat portion are movable axially relative to one another.
